ROBLOX STUDIO EXPERT SKILLS · TEAM SKILL 2 OF 14

Code Architecture

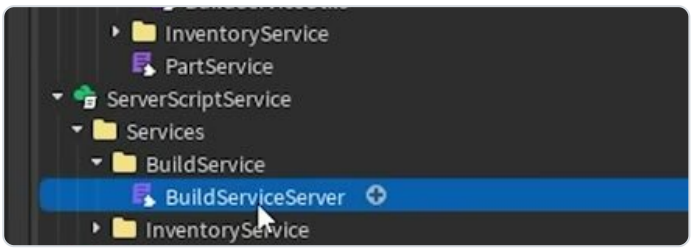
Gameplay programmer, game architect

`scenario-roblox-architecture`

Lays out Roblox game code so each script runs once on the right side, boots in two phases, and every connection is cleaned up, proven in live playtests.



Live playtest, Studio 0.743: HUD timer mirrored from server attributes, over the cloned map; server p95 17.3 ms.



One service split by side: client and utils replicated, server half in ServerScriptService. Leif, 00:05:09.

USE IT WHEN

- Where scripts go, client and server split, or picking a module loader or framework
- A script runs twice, or require fails with 'required recursively'
- A tool stops working after respawn, or memory climbs from leaked connections

WHAT THE EXPERTS INSIST ON

- No module may call another while loading: Init sets up its own state, Start may yield and call others. Suphi Kaner (Roblox scripting educator)
- A state service never requires its GUI: the GUI calls the service and listens. MrAsync (systems engineer, Super Social)
- Wire cleanup to the owner that dies with it, such as a per-tycoon Trove, not the player's. Crusherfire (Roblox scripter and educator)
- Exploiters can read ReplicatedStorage, so server modules never sit there; create remotes in code, never by hand. Leif (Roblox developer and educator)
- Knit-style frameworks decide for you and make everything depend on one service; use a class only for several stateful instances. B Ricey (Roblox educator)

WORKFLOW, WITH THE GATE BEFORE EACH NEXT STAGE

1. **Scaffold. Gate:** Server and Client RunContexts, Deferred signals; 0 type errors on both solvers.
2. **Map the modules. Gate:** ArchAudit: 0 errors, 0 warnings, no require cycle, 0 unresolved requires.
3. **Boot in two phases. Gate:** Both sides: BootOk, 0 errors, booted once; every init before every start.
4. **Mirror state as attributes. Gate:** Server state, client mirror and HUD text agree in a checked capture.
5. **Bind tags, own lifetimes. Gate:** Five respawns leave one character, one sword; Lobby holds 0 maps.
6. **Observe and iterate live. Gate:** Live audit tally matches offline; run_live.py exits 0.

KEY NUMBERS

Boot, 5 server services	**1.56 to 2.52 ms (live, Studio 0.743)**
Boot, 2 client controllers	**0.40 to 1.08 ms**
Five respawns mid-round	**instances within 3 of about 46,150; 1 character, 1 sword**
Map cycle, server frames	**p95 17.3 ms, worst 17.5 to 20.0 ms; Lobby delta 0**
Module size smell	**500 lines or more**
Player latency	**100 to 300 ms live; Studio playtests run at 0 ms**

CHECKS THE AGENT RUNS ON ITS OWN WORK

- ArchAudit 0 errors and 0 warnings outside fixtures; AgentKit Audit 0 errors.
- Boot report on both sides: BootOk true, BootErrors 0, BootCount 1.
- Console free of 'required recursively', '[Loader]' and 'non-legacy RunContext' lines.
- A round capture shows the HUD with mirrored state; image_checks raises no flag.

HANDOFFS

Receives Brief, toolkit and place slot from scenario-roblox-expert; tagged model templates from scenario-3d and scenario-game-assets.

Delivers The Rojo src tree, module map and ArchAudit report; service slots for networking, data, gameplay and UI.

TOOLKIT RBX_ARCHITECTURE.PY

- `scaffold()`: arena skeleton, Loader, Binder, Rojo project, built place
- `audit_file()`: offline ArchAudit: require cycles, unresolved requires, smells
- `boot_report()`: BootOk, errors and order on both sides

MAIN EXPERT SOURCES

MrAsync
systems engineer at Super Social, Writing Maintainable Codebases, Connect 2022

Suphi Kaner
Roblox scripting educator (unverified), How To Structure Your Project And Organize Your Scripts

Crusherfire
Roblox scripter and educator (unverified), These Tiny Mistakes Destroy Roblox Games (Memory Leaks)

Leif
Roblox developer and educator (unverified), How PRO Devs Set Up Roblox Games

STATUS

Tests 8 offline checks pass

In app 14 live tests pass in Roblox Studio 0.743

G1 code architecture

without skills		10/15
with skills		15/15